UNIT 1 · 8 hours

Signal Flow Graphs and Mason's Gain Formula



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
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Learning objectives

- Define the terms of a signal flow graph
- Draw an SFG from equations or a block diagram
- Identify forward paths, loops and non-touching loops
- Apply Mason's gain formula

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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Signal flow graph (SFG) terminology

Node and branch

Node = variable; branch = directed line with a gain (transmittance). Input node (source), output node (sink), mixed node.

Path and forward path

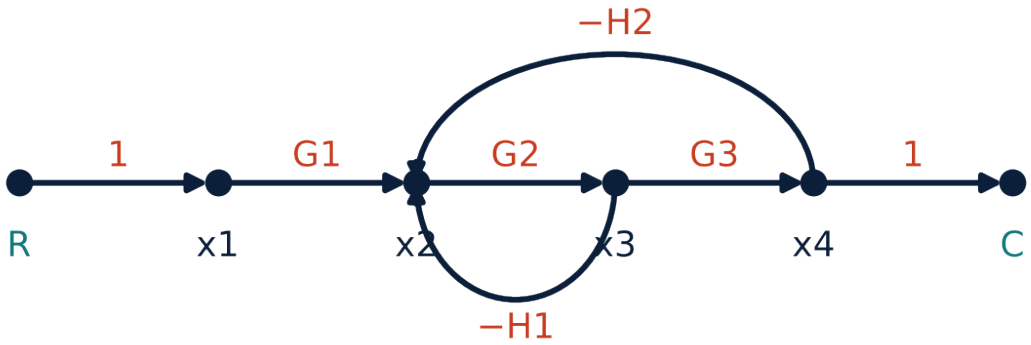
A path follows branch directions without repeating a node. Forward path: from input to output.

Loop

A closed path that returns to the same node; loop gain = product of branch gains. Non-touching loops share no node.

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Example SFG



- Nodes: R, x1, x2, x3, x4, C.
- Forward path: R→x1→x2→x3→x4→C.
- Loop L1: x2→x3→x2 with gain −G2H1.
- Loop L2: x2→x3→x4→x2 with gain −G2G3H2.

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Mason's gain formula

$$T = C/R = \Sigma P_k \Delta_k / \Delta$$

overall gain

$$\Delta = 1 - \Sigma L_i + \Sigma L_i L_j - \Sigma L_i L_j L_k + \dots$$

graph determinant

$$\Delta_k$$

Δ with all loops touching path k removed

- Pk = gain of the k-th forward path.
- Li = loop gains; LiLj = products of gains of non-touching pairs, etc.

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Applying Mason's formula to the example

Problem Find C/R for the SFG with forward gain 1·G1·G2·G3·1 and loops −G2H1 and −G2G3H2.

- P1 = G1 G2 G3
- L1 = −G2H1, L2 = −G2G3H2 (touching)
- Δ = 1 − (L1 + L2) = 1 + G2H1 + G2G3H2
- Δ1 = 1 (no loop is disjoint from P1)

Answer: C/R = G1G2G3 / (1 + G2H1 + G2G3H2)

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Numerical: with non-touching loops

Problem Loop gains L1 = −0.2, L2 = −0.5, L3 = −0.1; only L1 and L3 are non-touching. One forward path P1 = 5 touches all loops. Find T.

- $\Sigma L_i = -0.8$
- ΣL_iL_j (non-touching) = $L_1L_3 = (-0.2)(-0.1) = 0.02$
- $\Delta = 1 - (-0.8) + 0.02 = 1.82;\ \Delta_1 = 1$
- $T = P_1\Delta_1/\Delta = 5/1.82$

Answer: T = 2.75

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Block diagram to SFG

1

Nodes

Draw a node for every signal (input, output, each summing/take-off point output).

2

Branches

Each block becomes a branch with the block's gain.

3

Summing points

Branch gains ±1 into the summing node.

4

Apply Mason

Find paths, loops, non-touching loops and evaluate.

Remember: negative feedback appears as a loop with negative gain.

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Key takeaways

- SFG: nodes (variables) and directed branches (gains)
- Mason: $T = \Sigma P_k \Delta_k / \Delta$
- $\Delta = 1 - \Sigma L + \Sigma L_i L_j$ (non-touching) − …
- Δ_k excludes loops touching the k-th forward path

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Practice questions

Q1. Define forward path, loop and non-touching loops.

Q2. Write Mason's gain formula and explain each term.

Q3. Find T for P1 = 2, loops −1 and −0.5 (touching).

Q4. Convert a unity-feedback block diagram into an SFG.